

7B Week 8 Solutions

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1. The work needed is equal to the change in internal energy:

$$W = q\Delta V = (3C)(22V - 17V) = 15J. \quad (1)$$

2. (a) Usin law of cosines we find:

$$r_+ = \sqrt{r^2 + (d/2)^2 - 2r(d/2) \cos \theta} \quad (2)$$

$$= \sqrt{r^2(1 + (d/2r)^2 - 2(d/2r) \cos \theta)} \quad (3)$$

$$\approx r \left(1 - \frac{d}{r} \cos \theta\right)^{\frac{1}{2}}; \text{ where we discard } (d/2r)^2 \text{ term} \quad (4)$$

$$\approx r \left(1 - \frac{1}{2} \frac{d}{r} \cos \theta\right); \text{ where we've used Taylor expansion} \quad (5)$$

$$\boxed{= r - \frac{1}{2} d \cos \theta} \quad (6)$$

Comment: We could have also derived this by drawing a perpendicular line from $+q$ onto $\vec{r}$ and arguing that the bigger sub-triangle would be isosceles.

Similarly, we find $\boxed{r_- \approx r + \frac{1}{2} d \cos \theta}$ which we basically get from the previous part by changing $\theta \rightarrow \pi - \theta$.

(b)

$$V = V_{+q} + V_{-q} \quad (7)$$

$$= \frac{kq}{r_+} - \frac{kq}{r_-} \quad (8)$$

$$\approx kq \left(\frac{1}{r - \frac{1}{2} d \cos \theta} - \frac{1}{r + \frac{1}{2} d \cos \theta} \right) \quad (9)$$

$$= \frac{kq}{r} \left(\frac{1}{1 - \frac{1}{2} \frac{d}{r} \cos \theta} - \frac{1}{1 + \frac{1}{2} \frac{d}{r} \cos \theta} \right) \quad (10)$$

$$\approx \frac{kq}{r} \left(1 + \frac{1}{2} \frac{d}{r} \cos \theta - \left(1 - \frac{1}{2} \frac{d}{r} \cos \theta \right) \right) \quad (11)$$

$$= \frac{kqd \cos \theta}{r^2} \quad (12)$$

$$\boxed{= \frac{kp \cos \theta}{r^2}} \quad (13)$$

- (c) As the potential doesn't depend on ϕ , electric field won't have any ϕ component, and we can use the hint as:

$$\vec{E} = -\nabla V \quad (14)$$

$$= -\left(\partial_r \frac{kp \cos \theta}{r^2}\right) \hat{r} - \frac{1}{r} \left(\partial_\theta \frac{kp \cos \theta}{r^2}\right) \hat{\theta} \quad (15)$$

$$= \frac{2kp \cos \theta}{r^3} \hat{r} + \frac{kp \sin \theta}{r^3} \hat{\theta} \quad (16)$$

$$= \boxed{\frac{kp}{r^3} (2\hat{r} \cos \theta + \hat{\theta} \sin \theta)} \quad (17)$$

- (d) If P is on the line, then $r_+ = r - d/2$, $r_- = r + d/2$, and we can use the Coulomb's law:

$$\vec{E} = \vec{E}_+ + \vec{E}_- \quad (18)$$

$$= \left(\frac{kq}{r_+^2} - \frac{kq}{r_-^2}\right) \hat{r} \quad (19)$$

$$= \frac{kq}{r^2} \left(\frac{1}{\left(1 - \frac{1}{2} \frac{d}{r}\right)^2} - \frac{1}{\left(1 + \frac{1}{2} \frac{d}{r}\right)^2} \right) \hat{r} \quad (20)$$

$$= \frac{kq}{r^2} \left(1 + \frac{d}{r} - \left(1 - \frac{d}{r}\right) \right) \hat{r} \quad (21)$$

$$= \boxed{\frac{2kqd}{r^3} \hat{r}}, \quad (22)$$

which agrees with our result for $\theta = 0$.

- (e) If P is orthogonal $\theta = \pi/2$, then we will have (geometrical observation) that $\hat{r}$ components of $+q$ and $-q$ cancel, while $\hat{\theta}$ component is the same, and points in positive $\hat{\theta}$ direction:

$$\vec{E} = (E_{+, \theta} + E_{-, \theta}) \hat{\theta} \quad (23)$$

$$= 2 \frac{kq}{r_+^2} \sin(\alpha) \hat{\theta}; \text{ where } \sin \alpha = \frac{d/2}{r_+} \quad (24)$$

$$\approx 2 \frac{kq}{r^2} \frac{d/2}{r} \hat{\theta} \quad (25)$$

$$= \boxed{\frac{kqd}{r^3} \hat{\theta}}, \quad (26)$$

which agrees with our result for $\theta = \pi/2$.

3. Solution:

(a) The initial total energy is just the kinetic energy. There is no electric potential energy because the particle m is infinitely far away.

$$E_i = K$$

The final total energy is just the electric potential energy. There is no kinetic energy because the particle is at rest. Using $U=qV$,

$$V = kQ/r$$

$$E_f = qV = kQq/r$$

Using energy conservation,

$$K = kQq/r$$

$$r = kQq/K$$

(a)

(b) Initial energy is still $E_i = K$. We find the final energy as:

$$r' = 2r = 2 \frac{kQq}{K} \quad (27)$$

$$\Rightarrow qV' = \frac{kQq}{r'} = \frac{1}{2} \frac{kQq}{r} = \frac{1}{2} K \quad (28)$$

$$(\text{energy conservation}) \Rightarrow K' + qV' = K \quad (29)$$

$$\Rightarrow K' = \frac{1}{2} K \quad (30)$$

$$\Rightarrow \frac{1}{2} mv'^2 = \frac{1}{2} K \quad (31)$$

$$\Rightarrow \boxed{v' = \sqrt{K/m}}. \quad (32)$$

4. (a) Sketch.

(b) The total energy of a configuration of charges q_i where V_i is the potential from each charge is

$$U = \frac{1}{2} \sum_{i \neq j} q_i V_j$$

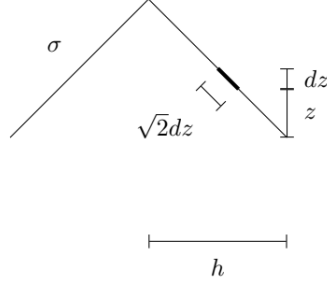
(c) Use $E = \sum_i E_i$.

(d) Use $V(P) = \sum_i V_i(P)$

(e) This is just $U = QV(P)$.

5. Parametrize using cylindrical coordinates

Consider a segment of the cone from z to $z + dz$ and from θ to $\theta + d\theta$. From the side, this looks as follows.



Since the side of the cone makes an 45° angle with the base, the diagonal length along the patch dq must be $\sqrt{2}dz$. The other direction is $r(z)d\theta = (h - z)d\theta$ since the radius of the circle at height z is $h - z$. Overall, then,

$$dq = \sigma(h - z) d\theta \sqrt{2} dz.$$

The distance from the base to dq is $\sqrt{z^2 + (h - z)^2}$ and the distance from dq to the top is $\sqrt{2}(h - z)$. Therefore the potential at the base is

$$V_{\text{base}} = \int_0^{2\pi} \int_0^h \frac{k\sqrt{2}\sigma(h - z)d\theta dz}{\sqrt{z^2 + (h - z)^2}} = \frac{1}{2}hk\sigma \ln \frac{1 + \sqrt{2}}{-1 + \sqrt{2}}$$

The potential at the peak is similarly

$$V_{\text{peak}} = \int_0^{2\pi} \int_0^h \frac{k\sqrt{2}\sigma(h - z)d\theta dz}{\sqrt{2}(h - z)} = hk\sigma,$$

so the difference in potential is

$$V_{\text{peak}} - V_{\text{base}} = hk\sigma - \frac{1}{2}hk\sigma \ln \frac{1 + \sqrt{2}}{-1 + \sqrt{2}} = \left(1 - \frac{1}{2} \ln \frac{1 + \sqrt{2}}{-1 + \sqrt{2}}\right) hk\sigma \approx 0.118hk\sigma.$$